

PEMBERTON, BC, Canada

WMO: 717770

Lat: 50.306N

Lon: 122.734W

Elev: 204

StdP: 98.89

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-11.6	-19.5	0.7	-11.4	-16.5	0.9	-9.2	6.2	-7.3	5.0	-6.7	1.8	270	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	33.6	18.9	31.4	18.1	29.1	17.3	20.5	29.8	19.1	28.5	18.1	26.9	2.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.8	23.2	16.1	11.7	21.2	15.0	10.9	19.8	59.9	29.2	55.2	28.7	51.7	26.9	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.1	4.6	3.5	-16.7	36.8	2.8	2.3	-18.7	38.5	-20.3	39.9	-21.9	41.2	-24.0	42.8
			-17.0	22.9	2.9	1.5	-19.1	24.0	-20.8	24.8	-22.4	25.7	-24.5	26.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-2.3	0.1	4.3	8.9	13.4	16.3	19.7	19.1	14.7	8.0	1.7	-2.3	(6)
(7)		DBStd	8.50	4.60	3.44	3.01	2.78	3.21	3.06	3.21	2.89	2.92	2.99	3.76	4.05	(7)
(8)		HDD10.0	1602	381	277	177	53	7	0	0	2	76	250	381		(8)
(9)		HDD18.3	3720	639	510	434	284	155	77	23	26	112	321	500	639	(9)
(10)		CDD10.0	1062	0	0	1	19	112	189	300	283	144	13	0	0	(10)
(11)		CDD18.3	139	0	0	0	0	2	16	65	51	4	0	0	0	(11)
(12)		CDH23.3	2489	0	0	0	14	153	362	1034	771	154	1	0	0	(12)
(13)		CDH26.7	1051	0	0	0	2	39	142	493	332	43	0	0	0	(13)
(14)	Wind	WSAvg	1.1	1.0	1.0	1.1	1.3	1.4	1.4	1.1	0.9	0.9	1.0	1.0	(14)	
(15)	Precipitation	PrecAvg	1022	142	90	87	52	43	49	34	47	59	140	155	143	(15)
(16)		PrecMax	1342	281	203	190	111	92	95	77	209	123	389	375	231	(16)
(17)		PrecMin	734	72	10	22	11	10	19	5	7	7	10	41	50	(17)
(18)		PrecStd	159	47	48	42	26	21	21	21	42	32	83	84	48	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	9.5	17.9	25.6	30.4	34.3	37.0	34.7	30.6	20.9	10.4	5.6	(19)
(20)		MCWB	3.7	5.3	9.2	13.0	15.9	19.3	20.4	19.4	17.5	13.3	9.4	4.8	(20)	
(21)		2%	DB	3.5	6.9	13.8	21.0	27.5	30.7	34.3	32.9	27.6	17.0	8.7	3.7	(21)
(22)		MCWB	3.0	4.5	7.6	11.2	14.8	17.5	19.2	18.9	16.8	11.0	7.9	3.4	(22)	
(23)		5%	DB	2.6	5.3	11.3	17.9	24.6	27.5	31.9	30.5	24.6	14.4	7.5	2.7	(23)
(24)		MCWB	2.3	3.5	6.5	9.6	13.8	16.4	18.4	18.1	15.7	10.8	6.9	2.4	(24)	
(25)		10%	DB	1.9	3.9	9.3	15.3	21.7	24.5	29.3	28.0	21.8	12.5	6.3	1.8	(25)
(26)		MCWB	1.7	2.7	5.7	8.5	12.7	15.1	17.7	17.5	14.7	10.2	5.8	1.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	6.6	10.5	13.8	17.1	22.6	23.6	21.7	19.0	14.6	9.6	4.9	(27)
(28)		MCDWB	4.3	8.2	15.9	23.1	26.9	29.4	31.5	30.7	27.1	19.3	10.1	5.1	(28)	
(29)		2%	WB	3.1	5.0	8.6	11.9	15.6	18.5	20.8	20.2	17.6	12.7	8.1	3.5	(29)
(30)		MCDWB	3.3	6.4	12.4	18.8	24.8	28.0	30.2	29.6	25.2	15.0	8.5	3.8	(30)	
(31)		5%	WB	2.3	3.8	7.3	10.6	14.4	16.9	19.3	19.0	16.4	11.4	7.0	2.5	(31)
(32)		MCDWB	2.5	4.8	10.3	16.2	22.5	25.7	29.3	27.9	23.0	13.4	7.4	2.7	(32)	
(33)		10%	WB	1.7	2.9	6.2	9.3	13.3	15.7	18.2	17.9	15.3	10.3	5.9	1.7	(33)
(34)		MCDWB	1.9	3.7	8.7	13.8	20.0	23.2	27.6	26.0	20.4	12.2	6.2	1.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	4.0	6.8	9.6	12.2	14.1	13.6	16.2	15.9	13.6	8.8	4.2	3.6	(35)
(36)		5% DB	MCDBR	3.4	8.3	13.5	18.3	20.9	21.1	22.3	21.7	19.8	12.9	4.8	3.4	(36)
(37)			MCWBR	3.1	5.9	7.7	8.7	9.2	9.0	8.5	9.3	10.2	7.9	3.9	3.0	(37)
(38)			MCDBR	3.2	6.9	11.7	15.7	17.8	18.7	20.8	19.5	17.7	11.0	4.6	3.3	(38)
(39)		5% WB	MCWBR	3.0	5.2	7.1	7.9	8.1	8.3	8.6	9.1	9.5	7.2	3.9	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.301	0.322	0.385	0.387	0.383	0.375	0.374	0.349	0.353	0.304	0.290	(40)	
(41)		taud	2.203	2.230	2.188	2.022	2.181	2.321	2.371	2.399	2.462	2.364	2.327	2.196	(41)	
(42)		Ebn at Noon	720	828	875	850	868	874	875	859	847	760	714	662	(42)	
(43)		Edh at Noon	74	93	117	155	139	122	114	106	90	83	66	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.86	1.73	2.53	4.08	5.35	5.25	5.85	5.12	3.52	1.97	0.96	0.70	(44)	
(45)		RadStd	0.15	0.26	0.41	0.52	0.52	0.64	0.59	0.38	0.31	0.32	0.19	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page